

Large language model

$$P(w_1, w_2, w_3, w_4, w_5)$$

$$= P(w_1) P(w_2 | w_1) P(w_3 | w_2, w_1) P(w_4 | w_{1:n-1})$$

Bayesian network

Domain knowledge

Objects → nodes

Relations → edges

Graph

Words

Linguistic relations

Categorical distribution

Transformer

$$P(w_k | w_{1:k-1})$$

$$P(w_k = j) = f_{(j)}(w_{1:k-1})$$

Bayesian network

nodes → Graph

edges → Graph

relations

Categorical distribution

Random variable

w_k

V

W

$1 \leq k \leq K$

$0 \leq v_k \leq 1$

$\sum_{k=1}^K v_k = 1$

$P(w = k)$

$= v_k$

Variational autoencoder (VAE) Bayesian network

Image generation

Semantics (people, chairs, ...)

Domain knowledge → Objects → nodes → Graph

Relations → edges

Images

Real Value

Semantics

Image

Multivariate Gaussian

Joint $p(X, Z)$

Multivariate Gaussian Distribution

Joint distribution

$p(X, Z) = p(Z) p(X|Z)$

$p(Z) = N(Z | \vec{\mu}, I)$

$p(X|Z) = N(X | f(Z), \text{diag}(g(Z)))$

$f(Z)$ $g(Z)$

$p(\vec{z}; \vec{\mu}, \Sigma)$

$= \frac{1}{(2\pi)^{\frac{n}{2}} |\Sigma|^{\frac{1}{2}}} \exp\left(-\frac{1}{2} (\vec{z} - \vec{\mu})^T \Sigma^{-1} (\vec{z} - \vec{\mu})\right)$

Representation

Sampling

Inference $p(Z|X)$

Learning

Data generation

Sampling

VAE network

$z_0 \sim p(z)$
 $x_0 \sim p(x|z_0) = N(x|f(z_0), \text{diag}(g(z_0)))$

Inference

$p(z|x)$
 $= \frac{p(z, x)}{p(x)}$

$p(x) = \int_z p(z, x) dz$

Variational inference

$p(z|x) \approx q(z)$

$p(z, x) = p(z) p(x|z)$
 $p(z) = N(z|\vec{0}, I)$
 $p(x|z) = N(x|f(z), \text{diag}(g(z)))$

Kullback Leibler (KL) divergence

$KL(q(z)||p(z|x))$

$= E_{q(z)} [\log \frac{q(z)}{p(z|x)}]$

for any value z ,
 $q(z) \approx p(z|x)$
 $\frac{q(z)}{p(z|x)} \approx 1$

Smaller KL $\rightarrow$ closer
 Representation
 Sampling
 Inference $p(z|x)$
 Learning

Approximation: minimize KL divergence

Optimization problem

$$\min_{\vec{u}, \Sigma} KL(q(z|\vec{u})||p(z|x))$$

u^*, Σ^*

$$\min_{\vec{u}, \Sigma} E_{q(z)}[\log q(z)] - E_{q(z)}[\log p(z, x)]$$

$$KL(q(z)||p(z|x))$$

$$= E_{q(z)}\left[\log \frac{q(z)}{p(z|x)}\right]$$

$$= E_{q(z)}[\log q(z) - \log p(z|x)]$$

$$= E_{q(z)}[\log q(z)] - E_{q(z)}[\log p(z|x)]$$

$$= E_{q(z)}[\log q(z)] - E_{q(z)}[\log p(z, x)] + E_{q(z)}[\log p(x)]$$

$$\begin{aligned} & \log p(z|x) \\ &= \log \frac{p(z, x)}{p(x)} \\ &= \log p(z, x) - \log p(x) \end{aligned}$$

$$\downarrow$$

$\log p(x)$ constant